

Operations Runbook

Pre-deployment Checks

Before deployment, verify that the target revision is known, database migrations are compatible, required environment variables are present, and the health endpoint can be reached. Production deployment should not proceed when a required secret, database connection, or dependency is unavailable.

Health Checks

The liveness check determines whether the application process is running. The readiness check verifies that required infrastructure is available before traffic is accepted. Readiness therefore provides a stronger signal than liveness for routing decisions.

Rate Limits

Application rate limits protect shared resources from unbounded request volume. Rate-limit decisions are enforced before expensive downstream work whenever possible. Redis-backed counters are used for distributed rate-limit state. A rejected request should not consume the downstream AI or document-processing capacity.

Timeouts and Retries

Timeouts define the maximum time an operation may consume before it is considered failed. Retries should be limited to operations that are safe to retry. Repeating a request after a timeout can amplify load if the original operation continues running, so retry policy must account for downstream capacity.

Capacity Baseline

The operational baseline records throughput, latency, error rate, concurrency, and resource saturation. A measured throughput increase is not automatically a capacity improvement when external dependencies or event-loop saturation are the limiting factors.

Monitoring

Operational monitoring should track request latency, HTTP error rates, database pool utilization, Redis availability, embedding failures, AI upstream failures, and streaming disconnects. Metrics should be correlated with deployment revision and request ID where available.

Incident Response

During an incident, first establish the affected service boundary and approximate start time. Preserve request IDs, deployment version, relevant metrics, and representative error messages. Avoid making unrelated configuration changes before the primary failure mode is identified.

Failure Handling

A failed document ingestion operation must not leave a partially indexed document that appears ready. Embedding failure is treated as an ingestion failure. Vector replacement is expected to be atomic so that an existing indexed document is not replaced by an incomplete set of chunks.

Rollback

Rollback should restore the last known-good application revision when a deployment introduces a confirmed application regression. Database rollback is a separate decision and must not be assumed to be safe merely because the application revision was rolled back.